

Dongsheng Ding

CONTACT INFORMATION	Department of Electrical Engineering and Computer Science University of Tennessee, Knoxville 1520 Middle Drive, Knoxville, TN 37996	Phone: (213) 574-9471 E-mail: dongshed@utk.edu URL: https://dongshed.github.io
TEACHING INTERESTS	I view teaching as a responsibility toward future generations. Technology rapidly advances, yet engineering course materials stay relatively static, limiting alignment with today's interdisciplinary innovations. My teaching aims to develop forward-thinking curricula that bridge interdisciplinary gaps and expand career possibilities.	
RESEARCH INTERESTS	Application of principles and tools in optimization and control to the <i>study of requirement-driven machine decision-making</i>. Topics of interest span: constrained reinforcement learning; multi-agent reinforcement learning; constrained generative modeling; constrained alignment of diffusion and large language models; robustness and resilience analysis of optimization algorithms; nonlinear control of fractional-order dynamical systems. Applications of interest include: autonomous robots/vehicles; generative artificial intelligence in applied science and engineering, such as natural language processing, image processing, and molecular design; decision-making systems for power grid, transportation, communication, and healthcare management.	
APPOINTMENT	Assistant Professor University of Tennessee, Knoxville Postdoctoral Researcher University of Pennsylvania Graduate Research Assistant University of Southern California Graduate Research Assistant & Teaching Assistant University of Minnesota, Twin Cities	August 2025 – Now September 2022 – July 2025 August 2017 – August 2022 August 2015 – August 2017
EDUCATION	PhD in Electrical Engineering , GPA: 4.0/4.0 University of Southern California Thesis: Provable reinforcement learning for constrained and multi-agent control systems Advisor: Mihailo R. Jovanović MS in Electrical Engineering , GPA: 3.9/4.0 University of Minnesota, Twin Cities ME in Control Theory & Engineering , GPA: 3.7/4.0 BE in Automation , GPA: 3.8/4.0 Zhejiang University, Hangzhou, China	August 2022 August 2017 March 2015 June 2011

SUBMITTED
PAPERS

1. D. Ding. “Convergence of natural policy gradient primal-dual methods for constrained convex MDPs,” 2026. (under review)
2. T. Wang, S. Li, Y. Sun, D. Ding, E. Dobriban. “Where to spend rollouts: Hit-utility optimal rollout allocation for group-based RLVR,” 2026. (under review)

REFEREED
PUBLICATIONS

Journals

1. D. Ding, K. Zhang, J. Duan, T. Başar, and M. R. Jovanović. “Convergence and sample complexity of natural policy gradient primal-dual methods for constrained MDPs,” *J. Mach. Learn. Res.*, 26(256) 1-76, 2025; also arXiv: 2206.02346.
2. Q. Wang, J. Zhang, D. Ding, and D. Qi, “Adaptive Mittag-Leffler stabilization of a class of fractional order uncertain nonlinear systems,” *Asian J. Control*, 18(6) 2343–2351, 2016.
3. D. Ding, D. Qi, and Q. Wang, “Asymptotic pseudo-state stabilization of uncertain fractional-order nonlinear systems with additive disturbance,” *Nonlinear Dyn.*, 81(1) 667–677, 2015.
4. Q. Wang, D. Ding, and D. Qi, “Mittag-Leffler synchronization of uncertain fractional order chaotic systems,” *Chinese Physics B*, 24(6), 2015.
5. D. Ding, D. Qi, and Q. Wang, “Nonlinear Mittag-Leffler stabilization of commensurate fractional-order nonlinear systems,” *IET Control Theory Appl.*, 9(5) 681–690, 2014.
6. D. Ding, D. Qi, X. Luo, J. Chen, X. Wang, and P. Du, “Convergence analysis and performance of an extended central force optimization algorithm,” *Appl. Math. Comput.*, 219(4), 2246–2259, 2012.
7. D. Ding, X. Luo, J. Chen, X. Wang, P. Du, and Y. Guo, “A convergence proof and parameter analysis of central force optimization algorithm,” *J. Convergence Inf. Technol.*, 6(10), 16–23, 2011.

Machine Learning and Artificial Intelligence Conferences

1. S. Khalafi, A. Ribeiro, and D. Ding*. “Unlearning in diffusion models: A unified framework with KL divergence and likelihood constraints,” in *Proceedings of the International Conference on Machine Learning*, Seoul, South Korea, 2026. (to appear, **acceptance rate** 26.6%, ***corresponding author**)
2. B. Zhang, S. Li, I. Hounie, O. Bastani, D. Ding*, A. Ribeiro. “Alignment of large language models with constrained learning,” in *Proceedings of the Advances in Neural Information Processing Systems*, San Diego, California, 2025; also arXiv:2505.19387. (**acceptance rate** 24.5%, ***corresponding author**)
3. S. Khalafi, I. Hounie, D. Ding*, A. Ribeiro. “Composition and alignment of diffusion models using constrained learning,” in *Proceedings of the Advances in Neural Information Processing Systems*, San Diego, California, 2025; also arXiv:2508.19104. (**acceptance rate** 24.5%, ***corresponding author**)

4. S. Rozada, D. Ding*, A. Marques, A. Ribeiro. “Deterministic policy gradient primal-dual methods for continuous-space constrained MDPs.” in *Proceedings of the AAAI Conference on Artificial Intelligence*, Philadelphia, Pennsylvania, 2025; also arXiv:2408.10015. (**acceptance rate 23.4%, *corresponding author**)
5. X. Huang, S. Li, E. Dobriban, O. Bastani, H. Hassani, and D. Ding*. “One-shot safety alignment for large language models via optimal dualization.” in *Proceedings of the Advances in Neural Information Processing Systems*, Vancouver, Canada, 2024; also arXiv: 2405.19544. (**acceptance rate 25.8%, spotlight, *corresponding author**)
6. S. Khalafi, D. Ding*, and A. Ribeiro. “Constrained diffusion models via dual training.” in *Proceedings of the Advances in Neural Information Processing Systems*, Vancouver, Canada, 2024; also arXiv:2408.15094. (**acceptance rate 25.8%, *corresponding author**)
7. D. Ding, Z. Huan, and A. Ribeiro. “Resilient constrained reinforcement learning.” in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, Valencia, Spain, 2024; also arXiv:2312.17194. (**acceptance rate 27.6%**)
8. D. Ding, C.-Y. Wei, K. Zhang, and A. Ribeiro. “Last-iterate convergent policy gradient primal-dual methods for constrained MDPs,” in *Proceedings of the Advances in Neural Information Processing Systems*, New Orleans, Louisiana, 2023; also arXiv:2306.11700. (**acceptance rate 26.1%**)
9. D. Ding, X. Wei, Z. Yang, Z. Wang, and M. R. Jovanović. “Sample efficient generalized Lagrangian policy optimization for safe multi-agent reinforcement learning,” in *Proceedings of the Learning for Dynamics and Control Conference*, Philadelphia, Pennsylvania, 2023; also arXiv:2306.00212.
10. D. Ding, C.-Y. Wei, K. Zhang, and M. R. Jovanović. “Independent policy gradient for large-scale Markov potential games: sharper rates, function approximation, and game-agnostic convergence,” in *Proceedings of the 39th International Conference on Machine Learning*, Baltimore, Maryland, 2022; also arXiv:2202.04129. (**acceptance rate 21.5%, 118/1117 long presentations**)
11. D. Ding, X. Wei, Z. Yang, Z. Wang, and M. R. Jovanović. “Provably efficient safe exploration via primal-dual policy optimization,” in *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, Virtual, 2021; also arXiv:2003.00534. (**acceptance rate 30%, 48/455 orals**)
12. D. Ding, K. Zhang, T. Başar, and M. R. Jovanović. “Natural policy gradient primal-dual method for constrained Markov decision processes,” in *Proceedings of the Advances in Neural Information Processing Systems*, Virtual, 2020. (**acceptance rate 20%**)
13. D. Ding, X. Wei, Z. Yang, Z. Wang, and M. R. Jovanović. “Fast multi-agent temporal-difference learning via homotopy stochastic primal-dual method,” in *the Optimization Foundations for Reinforcement Learning Workshop at NeurIPS*, Vancouver, Canada, 2019; also arXiv:1908.02805.

Control Conferences

1. D. Ding and M. R. Jovanović. “Policy gradient primal-dual mirror descent for constrained MDPs with large state spaces,” in *Proceedings of the 61st IEEE Conference on Decision and Control*, Cancún, Mexico, 2022.
2. D. Ding, K. Zhang, T. Basar and M. R. Jovanović, “Convergence and optimality of policy gradient primal-dual method for constrained Markov decision processes,” in *Proceedings of the 2022 American Control Conference*, Atlanta, Georgia, 2022.
3. D. Ding, X. Wei, H. Yu, and M. R. Jovanović. “Byzantine-resilient distributed learning under constraints,” in *Proceedings of the 2021 American Control Conference*, Virtual, 2021.
4. D. Ding, J. Yuan, and M. R. Jovanović. “Discounted online Newton method for time-varying time series prediction,” in *Proceedings of the 2021 American Control Conference*, Virtual, 2021.
5. D. Ding and M. R. Jovanović. “Global exponential stability of primal-dual gradient flow dynamics based on the proximal augmented Lagrangian: A Lyapunov-based approach,” in *Proceedings of the 59th IEEE Conference on Decision and Control*, Virtual, 2020.
6. D. Ding, X. Wei, and M. R. Jovanović. “Distributed robust statistical learning: Byzantine mirror descent,” in *Proceedings of the 58th IEEE Conference on Decision and Control*, Nice, France, 2019.
7. D. Ding and M. R. Jovanović. “Global exponential stability of primal-dual gradient flow dynamics based on the proximal augmented Lagrangian,” in *Proceedings of the 2019 American Control Conference*, Philadelphia, Pennsylvania, 2019.
8. D. Ding, B. Hu, N. K. Dhingra, and M. R. Jovanović. “An exponentially convergent primal-dual algorithm for nonsmooth composite minimization,” in *Proceedings of the 57th IEEE Conference on Decision and Control*, Miami Beach, Florida, 2018.
9. D. Ding and M. R. Jovanović. “A primal-dual Laplacian gradient flow dynamics for distributed resource allocation problems,” in *Proceedings of the 2018 American Control Conference*, Milwaukee, Wisconsin, 2018.
10. D. Ding, D. Qi, and Q. Wang, “Adaptive Mittag-Leffler stabilization of commensurate fractional-order nonlinear systems,” in *Proceedings of the 53rd IEEE Conference on Decision and Control*, Los Angeles, California, 2014.
11. D. Ding, G. Zhang, D. Qi, and H. Zhang, “Strategy analysis of an evolutionary spectrum sensing game,” in *Proceedings of the Intelligent Computing and Applications (LSMS & ICSEE)*, Shanghai, China, 2014. (**nominate paper award**)
12. D. Ding, D. Qi, and Q. Wang, “Alternative LMI characterizations for fractional-order linear systems,” in *Proceedings of the 33rd Chinese Control Conference*, Nanjing, China, 2014.

13. D. Ding, D. Qi, and Q. Wang, “Fractional-order integral state space modeling and quasi state analysis via block operational matrix scheme,” in *Proceedings of the 26th Chinese Control and Decision Conference*, Changsha, China, 2014.

RESEARCH

Invited Talks & Posters

PRESENTATIONS

1. Invited talk of “Controllable generative diffusion models: A constrained optimization view,” in *the INFORMS 2026 Annual Meeting*, San Francisco, California, 2026.
2. Invited talk of “Constrained policy optimization: A tale of regularization and optimism,” in *the IFAC Workshop on Contextual Control: Integrating Learning, Optimization, and Physical Structures in Control Systems*, Busan, Republic of Korea, 2026.
3. Invited talk of “Controllable generative diffusion models: A constrained optimization view,” in *the Yusu Wang’s Group Seminar*, University of California, San Diego, California, 2026.
4. Invited talk of “Constrained policy optimization for large language model alignment,” in *the Statistics and Data Science Seminar*, University of Tennessee, Knoxville, 2025.
5. Invited talk of “Constrained policy optimization for large language model alignment,” in *the SIAM NNP Section Conference*, State College, Pennsylvania, 2025.
6. Invited talk of “Requirement-driven machine decision-making: From reinforcement learning to generative artificial intelligence,” in *the EECS Faculty Candidate Seminar*, University of Tennessee, Knoxville, 2025.
7. Invited talk of “Multi-agent reinforcement learning for large-scale Markov potential games,” in *the Workshop on Multi-Agent Learning in Dynamic Environments*, Texas A&M Institute of Data Science, TAMU, 2024.
8. Invited talk of “Multi-agent reinforcement learning for large-scale Markov potential games,” in *the ESE PhD Colloquium*, UPenn, 2024.
9. Invited talk of “Constrained policy optimization: A tale of regularization and optimism,” in *the INFORMS Optimization Society Conference*, Houston, Texas, 2024.
10. Invited talk of “Constrained policy optimization: A tale of regularization and optimism,” in *the ESE PhD Colloquium*, UPenn, 2023.
11. Invited talk of “Provable constrained policy optimization in reinforcement learning,” in *the Safe Reinforcement Learning Online Seminar*, Virtual, 2023.
12. Invited talk of “Finite-time performance of policy optimization methods for constrained reinforcement learning,” in *the INFORMS 2022 Annual Meeting*, Indianapolis, Indiana, 2022.
13. Invited poster of “Independent policy gradient for large-scale Markov potential games” in *the REAL@USC-Meta center workshop*, ECE, USC, 2022.

14. Invited poster of “Provably efficient safe exploration via primal-dual policy optimization” in *the 11th Annual Research Festival*, ECE, USC, 2021.
15. Invited talk of “Provable constrained policy optimization for reinforcement learning” in *the 38th Southern California Control Workshop*, University of California, Irvine, California, Virtual, 2021.
16. Invited poster of “Distributed robust statistical learning: Byzantine mirror descent” in *the 10th Annual Research Festival*, ECE, USC, 2019.
17. Invited poster of “An exponentially stable primal-dual algorithm for nonsmooth optimization” in *the 9th Annual Research Festival*, ECE, USC, 2018.
18. Invited talk of “A primal-dual algorithm for distributed resource allocation” in *the 34th Southern California Control Workshop*, University of California, Riverside, California, 2018.

Conference Talks & Posters

1. Contributed poster of “Composition and alignment of diffusion models using constrained learning” in *the 39th Conference on Neural Information Processing Systems*, San Diego, California, 2025.
2. Contributed poster of “Alignment of large language models with constrained learning” in *the 39th Conference on Neural Information Processing Systems*, San Diego, California, 2025.
3. Contributed poster of “Deterministic policy gradient primal-dual methods for continuous-space constrained MDPs” in *the 39th AAAI Conference on Artificial Intelligence*, Philadelphia, Pennsylvania, 2025.
4. Contributed poster of “Constrained diffusion models via dual training” in *the 38th Conference on Neural Information Processing Systems*, Vancouver, Canada, 2024.
5. Contributed poster of “One-shot safety alignment for large language models via optimal dualization.” in *the 38th Conference on Neural Information Processing Systems*, Vancouver, Canada, 2024.
6. Contributed talk & poster of “Last-iterate convergent policy gradient primal-dual methods for constrained MDPs” in *the 37th Conference on Neural Information Processing Systems*, New Orleans, Louisiana, 2023.
7. Contributed poster of “Sample efficient generalized Lagrangian policy optimization for safe MARL” in *the 5th Annual Learning for Dynamics and Control Conference*, Philadelphia, Pennsylvania, 2023.
8. Contributed talk of “Policy gradient primal-dual mirror descent for constrained MDPs with large state spaces,” in *the 61st IEEE Conference on Decision and Control*, Cancún, Mexico, 2022.
9. Contributed talk of “Policy gradient primal-dual method for constrained MDPs,” in *the 2022 American Control Conference*, Atlanta, Georgia, 2022.

10. Contributed talk & poster of “Independent policy gradient for large-scale Markov potential games: sharper rates, function approximation, and game-agnostic convergence” in *the 39th International Conference on Machine Learning*, Baltimore, Maryland, 2022. (118/1117 long presentations)
11. Contributed talk & poster of “Provably efficient safe exploration via primal-dual policy optimization” in *the 24th International Conference on Artificial Intelligence and Statistics*, Virtual, 2021. (48/455 orals)
12. Contributed talk & poster of “Natural Policy Gradient Primal-Dual Method for Constrained Markov Decision Processes” in *the 34th Conference on Neural Information Processing Systems*, Virtual, 2020.
13. Contributed talk of “Global exponential stability of primal-dual gradient flow dynamics based on the proximal augmented Lagrangian” in *the 59th IEEE Conference on Decision and Control*, Virtual, 2020.
14. Contributed poster of “Fast multi-agent temporal-difference learning via homotopy stochastic primal-dual method,” in *the Optimization Foundations for Reinforcement Learning Workshop at NeurIPS*, Vancouver, Canada, 2019; *the Southern California Machine Learning Symposium*, UCSD, 2020.
15. Contributed talk of “Exponential stability of primal-dual gradient flow dynamics based on proximal augmented Lagrangian,” in *the 2019 American Control Conference*, Philadelphia, Pennsylvania, 2019.
16. Contributed talk of “Nonsmooth composite minimization: an exponentially convergent primal-dual algorithm,” in *the 57th IEEE Conference on Decision and Control*, Miami Beach, Florida, 2018.
17. Contributed talk of “A primal-dual Laplacian gradient flow dynamics for distributed resource allocation problems,” in *the 2018 American Control Conference*, Milwaukee, Wisconsin, 2018.
18. Contributed talk of “Adaptive Mittag-Leffler stabilization of commensurate fractional-order nonlinear systems” in *the 53rd IEEE Conference on Decision and Control*, Los Angeles, California, 2014.
19. Contributed talk of “Alternative LMI characterizations for fractional-order linear systems” in *the 33rd Chinese Control Conference*, Nanjing, China, 2014.
20. Contributed talk of “Fractional-order integral state space modeling” in *the 26th Chinese Control and Decision Conference*, Changsha, China, 2014.

HONORS & AWARDS

Generative Artificial Intelligence (GenAI) Faculty Fellowship , The University of Tennessee, Knoxville	2026
Sigma Xi Membership , Sigma Xi, The Scientific Research Honor Society	2026
Scholar Award , Conference on Neural Information Processing Systems	2023
Expert Reviewers , International Conference on Machine Learning	2021
Travel Award , Conference on Neural Information Processing Systems	2020
Top Reviewers , International Conference on Machine Learning	2020

Travel Award , IEEE Conference on Decision and Control	2020
Travel Award , American Control Conference	2018, 2019, 2022
MHI PhD Scholar Finalist , ECE, University of Southern California	2018, 2021
ECE Department Fellowship , University of Minnesota	2015
Honor for Outstanding Graduate Student , Zhejiang University	2015
Nominate Paper Award , LSMS & ICSEE, 2014, Shanghai	2014
Bosch Scholarship , Bosch in China	2013
The First-Class of Graduate Scholarship , Zhejiang University	2012–2015
National Scholarship , Ministry of Education of P.R. China	2011
Wei Shaoxiang Engineering Talent , Wei Shaoxiang Foundation, HongKong	2010
The Second-Class of Physics and Technology Innovation Contest , Zhejiang Physical Society, Zhejiang, China	2009
The First-Class of Advanced Mathematics Contest , Zhejiang Mathematical Society, Zhejiang, China	2008

ACADEMIC SERVICE

Journal Referee

IEEE Transactions on Control of Network Systems
 IEEE Transactions on Automatic Control
 IEEE Robotics and Automation Letters
 IEEE Transactions on Intelligent Vehicles
 IEEE Control Systems Letters
 Systems & Control Letters
 Optimization Letters
 Machine Learning
 Neural Networks
 Automatica
 IEEE Transactions on Pattern Analysis and Machine Intelligence
 APSIPA Transactions on Signal and Information Processing
 Engineering Applications of Artificial Intelligence
 SIAM Journal on Mathematics of Data Science
 Transactions on Machine Learning Research
 Journal of Machine Learning Research
 IET Control Theory & Applications
 International Journal of Robust and Nonlinear Control
 Frontiers of Information Technology & Electronic Engineering
 International Journal of Systems Science
 The Journal of the Franklin Institute
 Journal of Applied Mathematics and Computing
 Nonlinear Dynamics
 IEEE Access

Conference Referee

IEEE Conference on Decision and Control	2018–2026
Conference on Neural Information Processing Systems	2020–2026

International Conference on Learning Representations	2021–2026
International Conference on Artificial Intelligence and Statistics	2021, 2025, 2026
International Conference on Machine Learning	2020–2026
ICML Workshop on Next Generation of AI Safety	2024
ICML Workshop on Theoretical Foundations of Foundation Models	2024
NeurIPS Safe Generative AI Workshop	2025
Learning for Dynamics and Control Conference	2024, 2025
AAAI Conference on Artificial Intelligence	2023, 2024, 2026
ACM Urban-AI Workshop	2026
American Control Conference	2018–2025
IFAC World Congress	2020
Chinese Control Conference	2014
Chinese Control and Decision Conference	2014

Conference Volunteer

38th International Conference on Machine Learning, Virtual	2021, 2022
24th International Conference on Artificial Intelligence and Statistics, Virtual	2021
35th Conference on Neural Information Processing Systems, Virtual	2021

Area Chair of NeurIPS Workshop

Workshop on Constrained Optimization for Machine Learning, San Diego	2025
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Co-chair of Nonlinear System and Control Section

26th Chinese Control and Decision Conference, Changsha, China	2014
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Admissions Committee, University of Pennsylvania

PhD Admission in the Department of Electrical and Systems Engineering	Fall 2023
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TEACHING

Instructor, University of Tennessee, Knoxville

COSC 594 <i>Interplay between Reinforcement Learning and Generative Artificial Intelligence</i>	Fall 2025
ECE 315 <i>Signals and Systems</i>	Spring 2026
COSC 594 <i>Optimization for Machine Learning</i>	Fall 2026

Lab Assistant, University of Pennsylvania

ESE 2000 <i>Artificial Intelligence Lab: Data, Systems, and Decisions</i>	Summer 2024
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Guest Lecturer, University of Pennsylvania

ESE 5140 <i>Graph Neural Networks</i>	Fall 2023
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Teaching Assistant, University of Southern California

EE 587 <i>Nonlinear Systems</i>	Spring 2018
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Guest Lecturer, University of Minnesota Twin Cities

EE 3015 <i>Statistical Methods in Electrical and Computer Engineering</i>	Spring 2017
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Teaching Assistant, University of Minnesota Twin Cities

EE 4231 <i>Automatic Control Systems</i>	Fall 2016
EE 3015 <i>Statistical Methods in Electrical and Computer Engineering</i>	Spring 2017
EE 8231 <i>Optimization Theory</i>	Spring 2017

ADVISING & MENTORING	PhD Committee Member , University of Tennessee, Knoxville	
	Fanqi Wang, Electrical Engineering and Computer Science	December 2025
	Mentor for Doctoral Students , University of Pennsylvania	
	Shervin Khalafi (2nd year)	September 2023 – Now
	Topic: <i>Constrained learning for diffusion models</i>	
	Ignacio Hounie	September 2024 – Fall 2025
	PhD Thesis: <i>Learning with pointwise constraints</i>	
	Now: TBD	
	Xinmeng Huang	February 2024 – February 2025
	PhD Thesis: <i>Topics in statistical machine learning: multitask learning, uncertainty quantification, and language model alignment</i>	
	Now: <i>Quantitative Researcher, Citadel Securities</i>	
	Mentor for Doctoral Students , King Juan Carlos University, Madrid, Spain	
	Sergio Rozada	February 2024 – February 2025
	PhD Thesis: <i>Low-rank methods in reinforcement learning</i>	
	Now: <i>Assistant Professor, King Juan Carlos University (Spain)</i>	
	Mentor for Master Students , University of Pennsylvania	
	Zhengyan Huan	September 2023 – February 2024
	Topic: Resilient constrained reinforcement learning	
	Now: <i>Doctoral Student, Tufts University</i>	
	Mentor for Undergraduate Students , University of Southern California	
	Terry Lu	January 2022 – June 2022
	Now: <i>Master in Finance, Princeton University</i>	
	Academic Mentor for Master Students , University of Southern California	
	Viterbi Graduate Mentorship Program	August 2018 – June 2021
SKILLS	Python, L ^A T _E X, C/C++, Matlab, SQL	
MEMBERSHIPS	The Institute of Electrical and Electronics Engineers (IEEE)	
	IEEE Control Systems Society Membership	
	Sigma Xi	
	The Institute for Operations Research and the Management Sciences (INFORMS)	
	Stanford Encyclopedia of Philosophy	
GRADUATE COURSE HIGHLIGHTS		
	Control Systems: EE 5231 <i>Linear System and Optimal Control</i> , EE 8215 <i>Nonlinear Systems</i> , AEM 8421 <i>Robust Multi-Variable Control Systems</i> , AEM 8423 <i>Convex Optimization Methods in Control</i> ; Optimization and Computation: EE 5239 <i>Introduction to Nonlinear Optimization</i> , EE 8231 <i>Optimization Theory</i> , ISE 633 <i>Large-Scale Optimization for Machine Learning</i> , CSCI 5304 <i>Computational Aspects of Matrix Theory</i> , CSCI 8314 <i>Sparse Matrix Computations</i> ; Probability and Machine Learn-	

ing: MATH 507A/B *Theory of Probability*, EE 556 *Stochastic Systems & Reinforcement Learning*, CSCI 5525 *Machine Learning*, EE 546 *Mathematics of High-Dimensional Data*, DSO 699 *Statistical Learning Theory*, CSCI 699 *Theoretical Machine Learning*, EE 5581 *Information Theory and Coding*.